

Surya Teja Anupindi

Machine Learning Engineer

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Machine Learning Engineer with a Master's in Data Science and hands-on experience building end-to-end ML pipelines across healthcare, environmental science, and logistics domains. Proficient in Python, PyTorch, TensorFlow, and Scikit-learn with expertise in NLP, Computer Vision, and multimodal deep learning.

RELEVANT WORK EXPERIENCE

Quantela, Hyderabad, TG

Apr 2022 – May 2024

Machine Learning Engineer

- Architected end-to-end MLOps pipelines (ingestion → training → deployment → monitoring) on AWS/GCP, reducing model deployment time from 6 weeks to under 48 hours and cutting batch training cost by ~45%.
- Pioneered an LLM-powered citizen-service automation workflow integrated into the eGovernance stack, resolving 89% of 40K+ monthly queries without human intervention and reducing response time from 14 min to under 30 sec.
- Established model governance (drift detection, SHAP-based explainability, A/B testing) aligned to CMMI Level 5, passing external AI audits with zero critical findings.

Quantela, Hyderabad, TG

2021 – Apr 2022

Machine Learning Intern

- Productionized computer vision and anomaly-detection models for video monitoring, crowd management, and IoT telemetry across 35,000+ smart-streetlight and parking assets, achieving 94% mAP and a 62% drop in false-positive alerts.
- Built a reusable AI model store and feature-engineering framework that eliminated ~70% of duplicate modeling effort across Asset Management, Location Insights, and Citizen App product lines.

PROJECTS

ArgRAG: Argument-Aware Retrieval-Augmented Generation

Research Paper, 2025

Python, PyTorch, Transformers

- Improved factual accuracy by 3.4% and explanation quality by 21% (human-rated) by designing ArgRAG, a modular RAG pipeline that inserts an explicit argumentation layer between retrieval and generation for explainable NLP.
- Reduced retrieval variance by ensembling BM25, DPR, ColBERT, and SPLADE retrievers via Reciprocal Rank Fusion, achieving +4.4% Verdict EM over the best single retriever on the BEIR-FEVER benchmark.
- Fine-tuned a DeBERTa-v3 stance classifier to categorize retrieved passages, grounding LLM reasoning templates and reducing expected calibration error (ECE) to 0.043.

Multimodal Clinical Outcome Prediction – MIMIC-IV

2025

Python, PyTorch, SQLAlchemy

- Built deep fusion architectures combining structured tabular EHR data (demographics, vital signs, lab statistics) with unstructured clinician admit notes.
- Encoded clinical notes using BioClinicalBERT embeddings and optimized patient mortality prediction, achieving a +7% F1-score improvement over baseline models.

PROJECTS (CONTINUED)

Pathway of the Silicon Mind: Agentic Ascension

July 2026 – Present

Python, NumPy, Pandas, Automation

- Ascending through a 10-sequence structured roadmap from Python foundations to self-evolving agentic codebases.
- **Sequence 9 (The Coder - Completed):** Built custom automation pipelines including file category organizers, CSV-to-JSON grade processors, and interactive ledger consoles.
- **Sequence 8 (The Data Sculptor - In Progress):** Implementing high-dimensional matrix transformations and exploratory data analysis (EDA) using NumPy, Pandas, and Matplotlib.

Anomaly Detection in Chest Radiographs

2024

Python, TensorFlow, Keras, OpenCV, Scikit-learn

- Productionized chest X-ray abnormality detection (pneumonia and tuberculosis) using transfer-learned ResNet50 and DenseNet121 architectures.
- Integrated Grad-CAM visualization heatmaps to provide clinical interpretability by highlighting localized pathological features for radiologists.

Species Distribution & Environmental Factor Analysis

2024

Python, Pandas, Scikit-learn, Matplotlib

- Modeled complex relationships between species distribution counts and geographic/meteorological variables using Scikit-learn regressors.
- Engineered features for spatial density correlation and cleaned dataset noise using IQR-based outlier filters, improving R-squared by 12%.

Asteroid Diameter Prediction via MLP Neural Network

2024

Python, TensorFlow, Scikit-learn, Pandas

- Built and optimized a Multi-Layer Perceptron (MLP) regressor on NASA's physical and orbital asteroid databases to estimate asteroid diameter.
- Handled multi-collinearity using feature selection and implemented dropout layers for regularization, achieving low MAE/MSE and high R-squared metrics.

Delivery Route Optimizer with Real-Time Constraints

2024

Python, NetworkX, NumPy, Pandas

- Developed a real-time shipping route simulator utilizing NetworkX graph objects, integrating dynamic edge weights representing traffic and time windows.
- Benchmarked Dijkstra, A*, and Christofides TSP heuristic algorithms to deliver optimal sub-second query latency and reduce fuel routing costs by 15%.

EDUCATION

Saint Peter's University, Jersey City, NJ

Feb 2026

Master of Science in Data Science

Gandhi Institute of Technology and Management, India

Aug 2023

Bachelor of Technology in Computer Science & Engineering

SKILLS

Technical Skills: Deep Learning (PyTorch, TensorFlow, Keras), NLP, Transformers (Hugging Face, BERT, RoBERTa, DeBERTa, BioClinicalBERT, FLAN-T5), RAG & IR (BM25, DPR, ColBERT, SPLADE, FAISS, RRF), Computer Vision (OpenCV, CNNs, Grad-CAM), MLOps, Feature Engineering

Languages & Tools: Python, SQL, R, Java, C, SQLAlchemy, Git, Anaconda, NLTK, SacreBLEU

